Project Planning & Management

StockWise AI – Retail Demand Forecasting & Sales Prediction

DEPI Machine Learning Capstone · Team of 5 · Version 1.1 · 31 October 2026

This document covers the four planning items that accompany the Project Proposal: the project plan (timeline, Gantt chart, milestones, deliverables and resources), task assignment and roles, the risk assessment and mitigation plan, and the key performance indicators (KPIs) we will use to judge success.

Scope reminder: Level 1 (the graded machine-learning core) is the priority and is targeted for a first working version by Friday 6 November 2026, as requested by the instructor. Levels 2–4 (inventory layer, dashboard with AI assistant, cloud deployment) are planned extensions, built only as time allows.

> **Version 1.1.** Updated after the first review meeting with the instructor (31 October 2026): exact lag days, the reasons behind our tools, three new risks (AI-assistant hallucination, simulation assumptions and late feedback), and KPIs linked to the assistant and simulation test designs. The changes are listed in Section 5; the full feedback log and our detailed responses are in the Literature Review document.

1. Project Plan

1.1 Timeline Overview

Phase	Dates	Focus	Status
Planning	31 Oct	Phase 1 documents: proposal, plan, roles, risks, KPIs, literature review, requirements	Required
Week 1	31 Oct – 6 Nov	Level 1 first working version: data, features, 5 models, MLflow	Required
Week 2	7 – 13 Nov	Residual analysis, tuning, final test score, README	Required
Week 3	14 – 20 Nov	Level 2 inventory layer; dashboard prototype	As time allows
Week 4	21 – 30 Nov	Level 3 dashboard + AI assistant; testing; code freeze	Partly as time allows
Final	1 – 4 Dec	Report, slides, infographic, rehearsal, presentation	Required

1.2 Gantt Chart

Bars show who owns each task and when. Hatched bars are extensions that are built only if Level 1 is finished; red diamonds mark the milestones in Section 1.4.

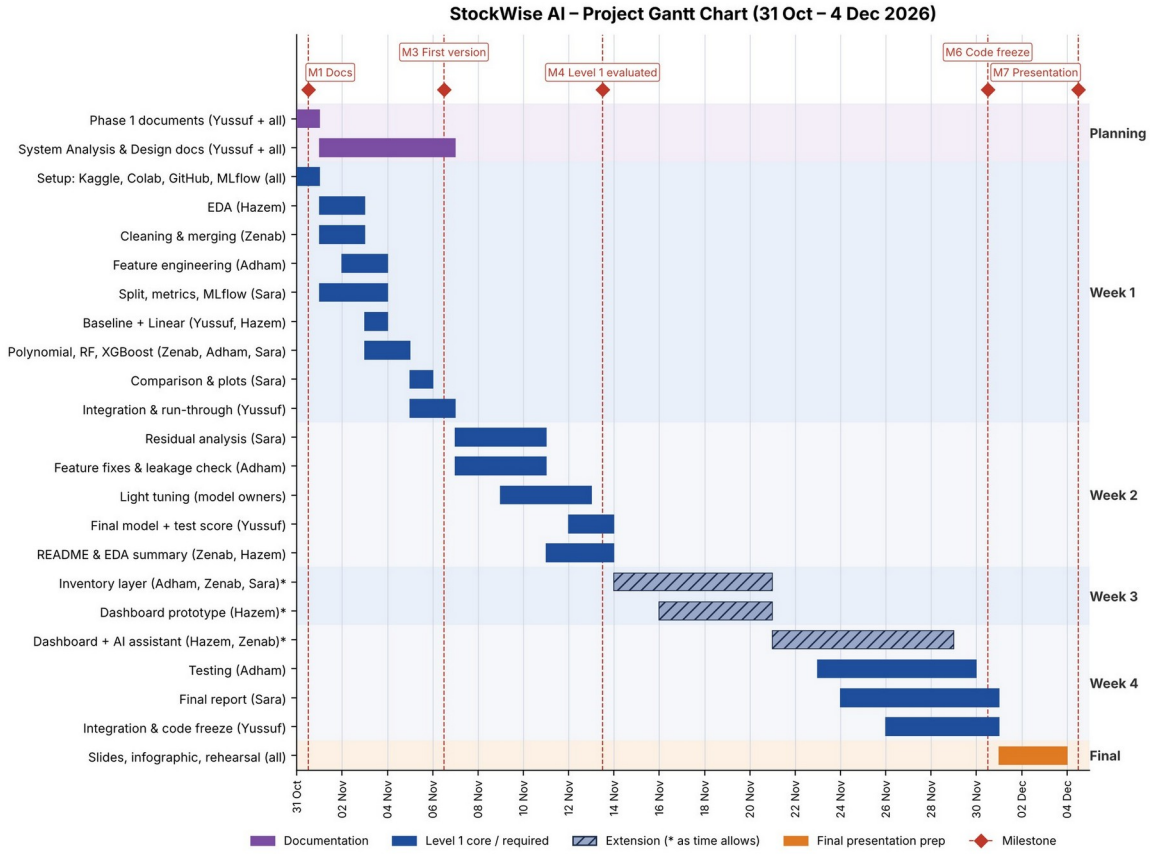


Figure 1. Project Gantt chart, 31 October – 4 December 2026

1.3 Week 1 Day-by-Day Plan

Week 1 is the most critical week because the instructor's target is a working Level 1 by Friday 6 Nov.

Day	Planned work
Sat 31 Oct	Submit Phase 1 documents. Everyone sets up Kaggle, Google Colab and GitHub and opens train.csv. Repository folders created. One MLflow test run logged.
Sun 1 Nov	EDA charts (Hazem). Merging stores, oil and holidays tables starts (Zenab). Date-based split and metric functions (Sara).
Mon 2 Nov	Cleaned dataset ready. Feature engineering: calendar, promotion and holiday features; lags of 16, 21 and 28 days; 7- and 28-day rolling means shifted by 16 days (Adham).
Tue 3 Nov	Feature table ready. Naive baseline (Yussuf) and Linear Regression (Hazem) logged in MLflow.
Wed 4 Nov	Polynomial Regression (Zenab), Random Forest (Adham) and XGBoost (Sara) logged in MLflow.
Thu 5 Nov	Model comparison table, predicted-vs-actual and first residual plots (Sara). Integration (Yussuf).
Fri 6 Nov	Short README and a full run-through of the pipeline to show the instructor.

1.4 Milestones

ID	Date	Milestone	Done when
M1	31 Oct	Phase 1 documents submitted	All documents uploaded to GitHub
M2	4 Nov	All five models trained	Five runs visible in MLflow
M3	6 Nov	Level 1 first working version	Pipeline runs end to end; comparison table ready
M4	13 Nov	Level 1 fully evaluated	Residual analysis, final test score, README
M5	20 Nov	Level 2 inventory layer (as time allows)	Reorder table per store × family
M6	30 Nov	Code freeze	Final code on GitHub; no new features
M7	4 Dec	Final presentation	Slides, report, infographic presented

1.5 Deliverables

Deliverable	Owner	Due
Phase 1 documents (proposal, plan, roles, risks, KPIs, literature review, requirements)	Yussuf (all contribute)	31 Oct
Literature Review updated with the instructor's feedback	Yussuf	After each review meeting
EDA notebook	Hazem	2 Nov
Cleaned dataset	Zenab	2 Nov
Feature-engineering pipeline (src/features.py)	Adham	3 Nov
Five models logged in MLflow	One model per member	4 Nov
Model comparison table and predicted-vs-actual plots	Sara	5 Nov
System Analysis & Design documents	Yussuf (all contribute)	6 Nov
Evaluation report with residual analysis	Sara	13 Nov
README	Zenab	13 Nov
Reorder recommendations and stock simulation (as time allows)	Adham, Zenab, Sara	20 Nov
Dashboard + AI assistant with guardrails (as time allows)	Hazem, Zenab	28 Nov
Test cases and results	Adham	29 Nov
Final source code on GitHub	Yussuf	30 Nov
Final report, slides, infographic	Sara (report), all	4 Dec

1.6 Resource Allocation

People. Five members. The workload is balanced by giving every member one pipeline stage and one model, so each person is busy throughout the week and can explain a model in the presentation. The table shows each member's main focus per week.

Member	Week 1	Week 2	Week 3	Week 4
Yussuf	Repo, integration, baseline	Final model, test score, code clean-up	Integrate inventory layer, architecture diagram	Final integration, code freeze
Hazem	EDA, Linear Regression	Tuning, EDA summary	Dashboard prototype*	Dashboard pages*
Zenab	Cleaning, Polynomial Regression	Tuning, README	Safety stock & reorder point*	AI assistant and guardrails*
Adham	Features, Random Forest	Feature fixes, leakage check	Simulated stock & lead times*	Testing
Sara	Split, metrics, MLflow, XGBoost	Residual analysis, tuning	Reorder flags*	Final report

** as time allows*

Tools and compute. All planned tools are free, and each was chosen for a specific reason:

- **Google Colab and Jupyter:** free, no installation, easy to share across the team. The free tier limits memory and session time (see risk R4).
- **pandas, NumPy, Matplotlib, Seaborn:** the standard Python data stack taught in the course, enough for about 3 million rows.
- **scikit-learn:** one interface for the Linear, Polynomial and Random Forest models; its Pipelines fit scalers on training data only, which prevents leakage.
- **XGBoost:** gradient-boosted trees performed strongly on retail sales data in forecasting competitions, train quickly on a CPU, and can be explained through feature importance.
- **MLflow:** required by the brief; records every run so results can be compared and reproduced. The tracking folder is on a shared Google Drive.
- **GitHub:** required by DEPI for code and documents; one branch per member keeps everyone's work safe.
- **Extensions:** Streamlit is planned for the dashboard because it is written in pure Python. The LLM provider and cloud platform are still to be decided, and free tiers will be preferred.

The full justification for each model and tool, with the alternatives considered and supporting references, is in the Literature Review, §2.1 and §3.

Time. Week 1 is planned day by day with a short daily check-in. From Week 2 the team holds one 30-minute sync per week (done / blocked / decisions) plus messages as needed.

2. Task Assignment & Roles

Every member writes code. Each member owns one stage of the pipeline and one model, reviews a partner's work, and must be able to explain their own part and the whole pipeline in the final presentation.

Member	Role	Responsibilities	Model	Reviews
Yussuf Akram	Team Leader & Integration	GitHub repository and folder structure, merging code into one pipeline, project documents and instructor feedback log, final model selection and test score, code freeze	Naive baseline	All merges into the main branch
Hazem Hussein	Data & EDA	Loading the data, time-oriented exploratory analysis (trends, seasonality, holidays, promotions, 2016 earthquake), EDA summary; later the dashboard*	Linear Regression	Zenab's cleaning
Zenab Yasser	Data Cleaning	Merging tables, oil-price gaps, transferred holidays, zero-sales series, outliers, cleaned dataset, README; later safety stock and the AI assistant with its guardrails*	Polynomial Regression	Hazem's EDA
Adham Ibrahim	Feature Engineering	Lags of 16, 21 and 28 days and 7- and 28-day rolling means shifted by 16 days, per store × family; calendar, promotion and holiday features; encoding (src/features.py); leakage check; testing; later simulated stock*	Random Forest	Sara's evaluation code
Sara Mahmoud	Evaluation & MLflow	Date-based split, metrics (MAE, RMSE, WAPE), MLflow setup, model comparison, residual analysis, plots, final report; later reorder flags*	XGBoost	Adham's features (leakage)

** as time allows*

Team Working Rules

- **Level 1 first:** nobody starts Level 2 or 3 until all Level 1 results are logged in MLflow.
- **GitHub:** one branch per member, clear commit messages, shared code in src/, only the team leader merges into main.
- **MLflow:** every training run is logged with its parameters and metrics, using one shared tracking folder.
- **Reviews:** each member's work is checked by their review partner before it is merged.
- **Understanding:** no copy-pasted code that the member cannot explain.
- **Instructor feedback:** after each review meeting, the feedback is logged in the Literature Review and applied to all documents that will be handed in.
- **Communication:** daily check-in in Week 1, then a weekly 30-minute sync; blockers are raised the same day.

3. Risk Assessment & Mitigation Plan

Likelihood and impact are rated Low, Medium or High. Each risk has one owner who watches for it. Risks R12–R14 were added in version 1.1.

ID	Risk	Likelihood	Impact	Mitigation	Owner
R1	One-week target for Level 1 is too tight	High	High	Simple models first; plan Week 1 day by day; daily check-in; use data from 2015 onward if needed; extensions only after Level 1	Yussuf
R2	First ML project: knowledge gaps slow the work	High	Medium	Small steps with mentor guidance; shared glossary; review pairs; each member learns their own stage deeply	All
R3	Data leakage gives results that look too good	Medium	High	Split by date only; lags of 16, 21 and 28 days only (never shorter than the 16-day horizon); shift before rolling; group by store × family; no transactions feature; leakage checklist review	Adham, Sara
R4	Colab runs out of memory or time on ~3M rows	Medium	Medium	Smaller data types; subset of recent years; sample for Random Forest and Polynomial; save intermediate files to Drive	Zenab
R5	Polynomial Regression creates too many features	Medium	Medium	Apply degree 2 to a few numeric features only	Zenab
R6	Messy data (oil gaps, transferred holidays, zero sales, 2016 earthquake)	High	Medium	Documented cleaning decisions; flag special periods; ignore transferred holidays	Zenab, Hazem
R7	A member becomes unavailable	Medium	Medium	Review partner knows the code; everything on GitHub; leader reassigns tasks	Yussuf
R8	Merge conflicts or work lost in notebooks	Medium	Medium	One branch per member; frequent commits; shared code moved to src/	Yussuf
R9	Scope creep: starting dashboard, LLM or cloud too early	Medium	High	Agreed rule: no Level 2 before Level 1 is logged in MLflow	Yussuf
R10	MLflow runs not shared or lost	Medium	Medium	One shared tracking folder; agreed run-naming convention	Sara
R11	API keys leaked or unexpected cost (Level 3)	Low	Medium	Keys kept in .env and .gitignore, never committed; free tiers preferred	Zenab
R12	AI assistant gives invented or wrong answers — hallucination (Level 3)	Medium	High	Tool-grounded design: the LLM only explains numbers returned by our tools; input validation; automatic number check before display; temperature 0; a person confirms every decision (guardrails G1–G7, Literature Review §2.3); tested with a prepared question set	Zenab

ID	Risk	Likelihood	Impact	Mitigation	Owner
R13	Simulation assumptions do not reflect reality (Level 2)	Medium	Medium	State every assumption; test lead times of 2, 4 and 7 days and service levels of 90%, 95% and 99%; take σ from real validation errors; confirm assumptions with the instructor (Literature Review §2.4)	Adham
R14	Instructor feedback arrives late and changes documents or scope	Medium	Medium	Log every meeting's feedback in the Literature Review; apply it to all documents within two days; keep a version history in each document	Yussuf

4. Key Performance Indicators (KPIs)

These KPIs are targets, not promised results: model accuracy is only known after training. All model KPIs are measured on the same test period (31 Jul – 15 Aug 2017). Comparing every model with a naive baseline, and using WAPE instead of percentage errors such as MAPE, follow standard forecasting practice; the reasons and references are in the Literature Review, §3.4.

4.1 Model Quality

KPI	Definition	Target	Measured
Beat the baseline	Best model's WAPE compared with the naive baseline's WAPE on the test period	Lower than baseline	13 Nov
Complete evaluation	MAE, RMSE and WAPE reported for all five models on the same test period	5 of 5 models	6 Nov
Residual analysis coverage	Errors analysed over time, by store, by product family, and holidays vs normal days	4 of 4 views	13 Nov
No data leakage	Issues found by the leakage checklist review in the final model	0 issues	13 Nov

4.2 Process & Delivery

KPI	Definition	Target	Measured
Experiment tracking	Training runs logged in MLflow with parameters and metrics	100% of runs	Weekly
Milestones on time	Milestones M1–M7 reached by their dates	7 of 7	Weekly
First version on time	Level 1 pipeline runs end to end	By 6 Nov	6 Nov
Reproducibility	A member who did not write a part can rerun the pipeline using only the README	Succeeds	13 Nov
Fair contribution	Each member commits their own work to GitHub every week	5 of 5 members	Weekly
Feedback applied	Instructor feedback points applied to all handed-in documents	All points	Within 2 days of each meeting
Presentation readiness	Members who answer examiner-style questions on their part and the pipeline in rehearsal	5 of 5 members	3 Dec

4.3 Extensions (as time allows)

KPI	Definition	Target	Measured
Reorder coverage	Store × family series with a reorder / OK / overstock recommendation	All 1,782 series	20 Nov
Simulation comparison	Stockout days and fill rate in the 16-day stock simulation, run with the naive-baseline forecast and with the best model (Literature Review §2.4)	Reported for both runs	20 Nov
Assistant correctness	Prepared test questions answered correctly from our own outputs (test design in Literature Review §2.3)	At least 8 of 10	28 Nov
Assistant refusals	Trick questions (a store that does not exist, a date outside the forecast window) answered with “no data” instead of a guess	All trick questions	28 Nov
Dashboard response time	Time to show the forecast for one store and product family	Under 5 seconds	28 Nov

5. Changes in Version 1.1

Changes made after review meeting 1 with the instructor (31 October 2026). Feedback IDs refer to the feedback log in the Literature Review.

Instructor feedback	Change in this document	Section
F1-01 Explain why each tool was chosen	Each tool now has its reason; the full justification is referenced	1.6
F1-02 State the lag days	Lags of 16, 21 and 28 days and 7- and 28-day rolling means shifted by 16 days stated in the Week 1 plan, roles and risk R3	1.3, 2, 3
F1-03 Prevent AI hallucination	New risk R12 with the guardrails as mitigation; assistant KPI linked to its test design; new refusal KPI	3, 4.3
F1-04 Safety stock in the simulation	New risk R13 on simulation assumptions; new simulation-comparison KPI	3, 4.3
F1-05 Stronger academic part	Tools and KPIs now refer to the research behind them (Literature Review §2.1, §3)	1.6, 4
F1-06 Apply feedback to all documents	Literature Review added to deliverables; new team rule, risk R14 and KPI for handling feedback	1.1, 1.5, 2, 3, 4.2